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ACUTE RHEUMATIC FEVER

Etiology : Post streptococcal disorder → **Group A β hemolytic streptococci**.

Age : School going children (**5-15 years**).

Incidence : m = F.

Pharyngitis $\xrightarrow[\text{molecular mimicry}]{10 \text{ days to few weeks}}$ Rheumatic fever

Pathogenesis :

Formation of Anti-streptococcal antibodies against Streptococcal antigens in the body $\longrightarrow$ Antibodies cross react with body's own tissues (Heart, joints and skin)

Probability $\begin{cases} \rightarrow 0.3\% \\ \rightarrow \text{In epidemic : 1-3\%} \end{cases}$

Risk stratification :

	Low risk population	moderate-high risk population (India)
Acute rheumatic fever : Incidence (Per lakh of school going children)	< 2	> 2
Rheumatic heart disease : Prevalence (Per 1000 population)	< 1	> 1

Clinical manifestations

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MODIFIED JONES CRITERIA (2015)

major criteria (moderate-high risk population) :

1. Pancarditis $\begin{cases} \rightarrow \text{Clinical} \\ \rightarrow \text{Subclinical (ECHO findings+)} \end{cases}$
 2. Arthritis : monoarthritis/polyarthritis/polyarthralgia.
 3. Subcutaneous nodules.
 4. Erythema marginatum.
 5. Sydenham's chorea
- Note : In low-risk population, **only polyarthritis** is considered a part of major criteria.

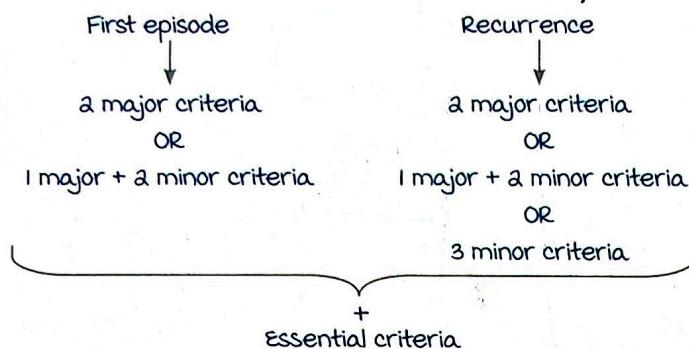
minor criteria :

	moderate-high risk population	Low risk population
Arthralgia	monoarthralgia.	Polyarthralgia
Fever	≥ 38 degree.	≥ 38.5 degree.
ESR $\&$ CRP	ESR ≥ 30 mm/hour $\&/$ or CRP ≥ 3 mg/dL.	ESR ≥ 60 mm/hour $\&/$ or CRP ≥ 3 mg/dL.
PR Interval	PR interval.	-

Essential criteria (Evidence of preceding streptococcal infection):

- **Elevated ASO titer** (Anti streptolysin O titer) : Preferred investigation.
- Throat swab culture (Risk of false positive/negative → not preferred).
- History of preceding sore throat (> 50 %).

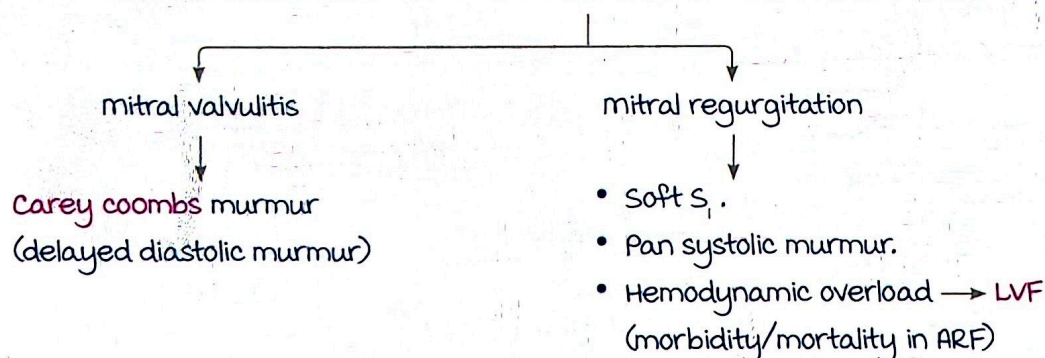
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DIAGNOSIS**Major Jones Criteria**

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Carditis :

- **m/c feature** of ARF (90%).
- Early finding (within 2 weeks).
- All 3 layers affected → Pancarditis.
- most serious : **Permanent damage** of valves can occur.
- Clinical diagnosis : khanirfan0392@gmail.com
 - Pericarditis : chest pain + frictional rub.
 - valvular involvement : m/c → mitral valve (F > m) > Aortic valve (m > F).

**Arthritis :**

- Large joints (knee, ankle, elbow) migratory arthritis.
- Early finding, 30-50% of affected children.
- Excellent response to **aspirin**.
- Complete recovery (No residual joint damage).

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Erythema marginatum :

- Rare, found in early stages.
- Lesion :
 - Erythematous with pale center → Coalesce → **Serpinginous rash**.
 - Non pruritic.
 - Starts in trunk & then involves limbs.
 - Face not involved.
 - Induced by warmth/heat.



Erythema marginatum

Subcutaneous Nodules :

- Late finding (6 weeks after onset).
- Painless nodules.
- Site : Around bony prominences (Elbow, shin, occiput, ankle)
- Associated with high risk of **carditis**.

Sydenham's Chorea :

- Last manifestation (3 months after onset).
- Semi purposeful, jerky movements of limbs → Abnormal coordination & posturing.
- Associated manifestations : Hypotonia, Emotional lability, Jerky speech.
- Signs :
 - **milkmaid grip sign** : Alternate tightening and loosening of grip.
 - **Darting tongue sign** : Uncontrollable sliding in & out of tongue on protrusion.
- Self resolving (2-6 weeks).
- Treatment (if not resolving) : Diazepam, Haloperidol.

Management

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Treatment & Prophylaxis :**Treatment :**

1. Bed rest for 2 weeks.
2. Penicillin :
 - Inj. Penicillin G (Benzathine) : Single dose. (or)
 - Oral Penicillin V 250 mg QID x 10 days. (or)
 - If allergic to Penicillin : Erythromycin 250mg QID x 10 days.
3. Anti inflammatory drugs : Steroids/aspirin.

2° Prophylaxis (To prevent further episodes/ carditis) :**Penicillin :****a. Benzathine/ Penicillin G :**

- > 30 kg : 12 lakh IU every 3 weeks.
- < 30 kg : 6 lakh IU every 2 weeks.

b. Oral Penicillin V : 250 mg BD.**c. If allergic to Penicillin : Erythromycin 250 mg BD.**

Acute rheumatic fever

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Anti inflammatory medications (For 12 weeks) :

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1. Steroids :

- Preferred (Especially in **cardiac** problems).
- Oral Prednisolone (2-60mg/kg/day) x3 weeks → Taper over 9 weeks.

2. Aspirin : 90-120 mg/kg/day in 4 divided doses x 10 weeks → Taper over next 2 weeks.

Duration of prophylaxis :

Scenario	Prophylaxis
No carditis	For next 5 years or till age 18, whichever is longer.
Carditis (without RHD)	For next 10 years or till age 25 (whichever is longer).
Established RHD/undergone surgery	Till age 40 (Ideal : Life long).

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